OVERVIEW OF THE COLLECTION

CLAUDE OPUS 5.5, GPT-6-ASTRA, CLAUDE FABLE 5.1

Abstract. This collection contains eight manuscripts, each resolving a published open question; four prove a conjecture or answer a question affirmatively, and four refute a conjecture. All main results are formally verified in Lean 4 with Mathlib. The manuscripts, Lean code and supporting computations were produced by AI models working as a team of agents; see the repository README for how the results were produced.

001. The maximum number of non-equivalent ancestral configurations

Discipline: mathematical biology (phylogenetics); enumerative combinatorics.

In the multispecies coalescent, gene tree probabilities are computed by dynamic programming over *ancestral configurations*. Disanto and Rosenberg (Bull. Math. Biol. 2019) introduced the number $c^*(T)$ of non-equivalent root configurations of a matching gene tree / species tree pair of shape T and conjectured that

$$\max_{|T|=n} c^*(T) = \left\lfloor 3^{\lfloor n/3 \rfloor}\, \tfrac{2r^2+8r+23}{27} \right\rfloor, \qquad r = n \bmod 3,\ n \geq 2,$$

having verified it for $n \leq 20$. The manuscript proves the conjecture for all n, shows that the maximizing shape is unique for $n \geq 9$ (it is obtained by repeatedly joining a three-leaf tree), determines the growth constant $3^{1/3}$ (previously known to lie in $[1.4422, 1.5028]$), and shows that the minimum is $n-1$, attained by exactly two shapes for $n \geq 4$. The key tool is the identity $s([L,R]) = 1 + s(L)s(R) - \binom{k}{2}$ for the number s of pruned shapes, with k the number of pruned shapes common to L and R. *Formalization:* two independent Lean proofs of the maximum, of uniqueness for $n \geq 9$ and of the minimum value.

002. The reversed-prime iteration of OEIS A100475 never enters a cycle

Discipline: elementary number theory.

OEIS A100475 iterates $k \mapsto \mathrm{rev}(p_k)$, where p_k is the k-th prime and rev reverses decimal digits, and asks whether a start other than 1 can lead to a loop; the question is an open statement in google-deepmind/formal-conjectures. The note shows that no start leads to a cycle: the number of digits never decreases along an orbit, $\pi(10^d) < 10^{d-1}$ for $d \geq 5$ confines any cycle to $[1, 1229]$, and a finite check excludes cycles there. In other bases cycles can exist: base 6 has the fixed point 11 and base 8 the fixed point 13. *Formalization:* a Lean proof of the negation of the formal-conjectures statement.

003. Maximally probable labeled topologies with simultaneity

Discipline: mathematical biology (phylogenetics); combinatorics of posets.

The tree U_n^* of Harding, Hammersley and Grimmett uniquely maximizes the number of labeled histories (orderings of the internal vertices compatible with ancestry) among binary trees with n leaves. Dickey and Rosenberg (Discrete Appl. Math. 2026) allowed several branching

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events to occur simultaneously and conjectured that U_n^* also uniquely maximizes the total number $N(T)$ of such tie-permitting histories, and maximizes the number $E(T, z)$ with z events for every z. The manuscript proves the conjecture in a stronger, coefficientwise form: for every bound m, the generating polynomial $\sum_f \prod_v x_{f(v)}$ of strict labelings of the internal vertices by $\{1, \ldots, m\}$ is maximized coefficientwise by U_n^*. The key tool is a switching injection comparing labelings of two trees with labelings of their union and intersection, with different label bounds on the two sides, used in an induction along the dyadic structure of U_n^*. *Formalization:* both parts of the conjecture including uniqueness, with the paper's recursive definition of U_n^*.

004. The minimum and maximum forcing numbers of bipartite graphs

Discipline: graph theory; mathematical chemistry (Kekulé structures).

The forcing number of a perfect matching M is the smallest number of edges of M that belong to no other perfect matching; in chemistry it is the "innate degree of freedom" of a Kekulé structure. Liu, Diwan and Zhang (arXiv 2025, Problem 5.5) asked whether every bipartite graph of order $2n$ whose maximum forcing number satisfies $F(G) \geq n-k$ has minimum forcing number $f(G) \geq \lceil n/k \rceil - 1$, and proved this for $k \leq 2$. The manuscript answers the question affirmatively for all k, in the equivalent form $n \leq (f(G)+1)(n - F(G))$, which is sharp. The key tools are the triangular structure of bipartite graphs with a unique perfect matching, an interval representation of a perfect matching, and the greedy algorithm for interval scheduling. *Formalization:* the statement of Problem 5.5 for all finite bipartite graphs with a perfect matching.

005. Minimum-variance phylogenetic trees are not unique

Discipline: mathematical biology (phylogenetics, tree balance).

Coronado, Mir, Rosselló and Rotger (BMC Bioinformatics 2020) studied the variance V of the leaf depths of a rooted bifurcating tree as a balance index and conjectured that, for every n, the trees with n leaves of minimum V all have the same multiset of leaf depths; they had checked this for $n \leq 2^{20}$. The manuscript disproves the conjecture: for $n = 67{,}256{,}579$ two trees with different leaf-depth multisets both attain the minimum $n^2V = 330{,}431{,}494{,}786$. Global minimality is proved by cutting a tree at a level, relaxing to an integer "coin" problem with a rigid normal form, and a finite certificate of 16,800 exact evaluations. Using the structure theorem of Coronado et al., an exact search shows that this is the smallest counterexample, and that there are exactly six counterexamples up to 2^{28}. *Formalization:* the negation of the conjecture, with the two trees and their global minimality among all trees.

006. Connected locally equienergetic graphs

Discipline: mathematical chemistry (graph energy); spectral graph theory.

The local energy of a graph is the sum over its vertices of the decrease of the energy (the sum of the absolute values of the adjacency eigenvalues) when the vertex is deleted. Dede and Popat (MATCH Commun. Math. Comput. Chem. 2025) stated, on the basis of a computer search, that up to order 10 the only connected pairs with equal local energy are (K_n, C_n) for odd n, and conjectured this for all orders. The manuscript shows that both statements are false: there are exactly 58 further pairs of order 10, all with the same eigenvalues and angles and related by Godsil–McKay switching; and at order 16 the rook's graph, the Shrikhande graph and the complement of the Clebsch graph all have local energy 64, although the last one has a different energy. Cospectral walk-regular graphs and strongly regular graphs with a

sign condition give infinite families of counterexamples. *Formalization:* the refutations by the pairs of orders 16 and 10, including a pair with different energies.

007. The two largest coset leaders modulo $(q+1)(q^m \pm 1)$

Discipline: coding theory (BCH codes); elementary number theory.

The largest q-cyclotomic coset leaders modulo n give the largest designed distances of BCH codes of length n. Liu, Wu and Zhu (arXiv 2026) conjectured, from computations, formulas for the two largest coset leaders modulo $(q+1)(q^m+1)$ and $(q+1)(q^m-1)$ for odd m. The manuscript proves both for all odd $m \geq 3$ (they fail for $m = 1$), the first for every odd $q \geq 3$ and the second for every $q \geq 2$. The key tool is a translation of cosets into orbits of signed or binary digit words under rotation, together with a classification of the words of small maximal rotation. *Formalization:* both conjectures for prime powers q and odd $m \geq 3$.

008. The third largest coset leader modulo $q^m + 1$

Discipline: coding theory (antiprimitive BCH codes).

Liu, Li, Guo and Song (Discrete Math. 2023, Conjecture 25) conjectured formulas for the second and third largest coset leaders modulo q^m+1 for q a power of 2. The manuscript shows that the formula for the third largest leader is false: for $q = 4$ and $m = 12$ the conjectured value is not even a coset leader. Computations locate the failures in the family $m = 3 \cdot 2^{r-1} \geq 12$ and suggest a corrected formula, which agrees with a theorem of Liu, Li, Fu, Lu and Rao for $q = 2$, $m = 12$. *Formalization:* the negation of the conjecture and the explicit counterexample.